

TXV IN EMERGENCY OPERATION



Typical symptom of TXV blockage /malfunction

* TXV check→ If there is frost around the TXV try to defrost with hot air gun(do not get overheated the TXV) or hot wet cloth ,warm the txv body (with hot water or something) and softly tap the TXV body. It could solve the problem for minor blockage.

***TXV: As a normal practice, TXV should not be adjusted as it needs correct super heat setting. But, if it is adjustable, should try if not feasible to replace new one.

→Please ensure to mark original setting of TXV before turning (adjust) in case you can set it back original setting.

TXV in Carrier units are not possible for adjustment. Most Thermoking TXV can be adjusted. Please check/adjust TXV in TK as below if not feasible to replace TXV.

Adjustment for TK TXV

→you will need imbus tool/allen key size:4mm. When adjusting you must remove cap from TXV. Turn(screw imbus fully "in" when turning "clockwise") for SHUT, compressor suction pressure will "decrease" till vacuum pressure.

When screw is fully-in then, unit must go in deep vacuum, when so, please open TXV counter clockwise 3.5 times to see if any suction pressure changes(should increase).

wait for 5 minutes to see suction pressure . Adjust to normal pressure around "0"Bar

Monitor evaporator and return temperature for improvement at the same time in controller display while adjusting. If no response at all, TXV might be defective or there will be other problem of Freon restriction.

TXV OPENING IN EMERGENCY

*If really found problem in TXV and not feasible to replace new TXV, it can be open/adjust by using tap screw or M6- bolt as below photo in emergency. But, it needs to be done with extra -care to avoid **puncturing of membrane** in TXV which will cause **heavy leak** of Freon if damage.

Please ensure to cut "sharp edge" before screwing/tapping. Tapping need to be done carefully and slowly. **DO NOT FORCE IT!!**

AA/ Method using Tapping bolt(recommended method)



1/ Cut away capillary tube of sensing bulb on top of TXV, then, the protruding part from the center of the txv-head need to be filed to achieve smooth surface.

2/ Use drill bit (2MM) size to drill a initial hole on TXV head, then, use 5MM size drill bit for final hole for M6 Bolt.

5mm bottom tap can be used for threading for the bolt .

The drilling need to be done carefully with the bits, not to drill more than (2mm) depth, take extra care to avoid membrane damage inside.

3/Use M6 bolt (8.8 tensile strength), (preferably Stainless steel) to tap slowly to adjust TXV.

4/ Monitor closely suction pressure , evaporator temperature, supply /return air while adjusting screw.

Subsequent manual adjustments of txv opening can be done based on pressure and temperature changes.

BB/ Method using Tapping screw(more risk for damaging TXV membrane)



1. Take a self cutting screw, cut the sharp-point off, so it will not make hole/puncture in membrane on TXV.

2. Cut capillary tube of the top of TXV valve.

3. Take the self cutting screw, and screw in hole from capillary tube. "Screwing-in" movement is adjusting opening on TXV valve. Need to check suction pressure while screwing-in. The suction pressure should be gradually increasing if TXV is opening. Also, check temperature improvement in controller display. The evaporator temperature and suction pressure are the key indications for TXV opening.
4. Adjust opening of TXV so unit will run on set-point, have to check adjustment several times a day.

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